

Tanish Baranwal

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Profile: UC Berkeley EECS student with research experience in machine learning and practical exposure to quantitative trading through internships, focusing on predictive modeling and systematic strategy development.

EDUCATION

University of California, Berkeley

Aug. 2022 – May 2026

Bachelor of Science in Electrical Engineering and Computer Sciences

Cumulative GPA: 3.98/4.0

- **Relevant Coursework:** Probability and Random Processes, Deep Neural Networks, Efficient Algorithms and Intractable Problems, Graduate Level Computer Vision, Operating Systems and System Programming, Physics-Inspired Machine Learning, AI for Biology and Chemistry, Algorithmic Aspects of Machine Learning
- **Organizations:** President (Spring 2025), Eta Kappa Nu (EECS Honor Society), Venture Strategy Solutions

EXPERIENCE

Berkeley Artificial Intelligence Research

Sep 2024 – Present

Undergraduate Researcher

Berkeley, CA

- Developed **Video Gaussian Masked Autoencoders**, a self-supervised video model that uses 3D Gaussian primitives to learn explicit spatiotemporal correspondences via masked reconstruction.
- Introduced a correspondence-aware pretraining strategy enabling **emergent zero-shot point tracking**, achieving state-of-the-art gains on TAP-Vid Kinetics (+**4.16%**), DAVIS (+**37.4%**), and Kubric (+**13.1%**).
- Advised by **Professor Jitendra Malik**; worked with Jathushan Rajasegaran and submitted to NeurIPS 2025.

GSA Capital

June 2025 – Sept 2025

Quantitative Research Intern

London, UK

- Developed an execution model in Jax to predict price improvement in broker SOR routing using signals and generalized linear models, reducing realized slippage by **15%** and informing broker SOR algorithm selection.
- Launched a systematic intraday strategy for London Metal Exchange futures from scratch, covering alpha research, risk-model hedging, and portfolio optimization, and deployed a live strategy with a backtested Sharpe of **1.9**.

Amazon

May 2024 – Aug 2024

Software Development Engineer Intern

Seattle, WA

- Improved product listing consistency on Amazon's storefront by building an LLM proof of concept that generated footwear size attributes for **100k+** products with **85%** accuracy using Bedrock and chain-of-thought prompting.
- Leveraged Amazon native AWS pipelines to automate the LLM agent to work with the catalog scale of **100 million** items using Lambda, S3, Apache Spark and AWS EMR, and Bedrock APIs and optimized for costs and efficiency.

eGolf.ai

Jan 2022 – Present

Co-Founder and Chief Technology Advisor

Saratoga, CA

- Designed an algorithm using Fourier Transform and CNNs to analyze golf swing data and predict statistics and used RANSAC, Kalman Filtering, and Pose2Sim to calculate athlete poses from video and motion sensors.
- Adapted CLAP foundation model with a Transformer decoder to predict temporal heatmaps for golf shot events, enabling fine-grained localization of ball impacts from noisy swing data, with **mean localization error of 0.2s**.

Cardiac Vision Labs, University of California, San Francisco

Sep 2022 – June 2025

Computer Vision Researcher

San Francisco, CA

- Designing novel diffusion-based generative models for cardiac excitation-wave simulation, achieving state-of-the-art performance in 3D wave imaging, arrhythmia diagnostics, and precise initial-condition modeling.
- Published a peer-reviewed paper on a state-of-the-art 3D cardiac depth-reconstruction model (**87% accuracy**) in AIP Journal of APL Machine Learning and presented at Computing in Cardiology 2023.

PUBLICATIONS

Baranwal, T., Lebert, J., & Christoph, J. (2024). Dreaming of Electrical Waves: Generative Modeling of Cardiac Excitation Waves using Diffusion Models. *APL Machine Learning*. <https://doi.org/10.1063/5.0194391>

Baranwal, T., Varada, S., Das, S., & Haider, M. R. (2024). Fault-Tolerant IoT System Using Software-Based "Digital Twin." *10th IEEE World Forum on Internet of Things*.

SKILLS AND INTERESTS

Skills: Python, C++, Matlab, Java, JAX, OpenCV, XGBoost, TensorFlow, PyTorch, Docker, AWS, MongoDB

Interests: Bodybuilding, Basketball, Football, Brooklyn 99, Swimming, Going to Concerts, Mixology, Baking